MIL-N-12314E
20 October 1983
SUPERSEDING
MIL-N-12314D
28 June 1977

MILITARY SPECIFICATION

NOZZLES, FIRE HOSE, WATER: SPRAY AND STREAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fire hose water nozzles for connecting to fire hose or playpipe.

1.2 Classification. The water nozzles shall be the following types, classes, and size, as specified (see 6.2):

Type I - Revolving, cellar, 2-1/2 inch.
Type II - Combination spray and stream.

Class A - Adjustable.

Size:

1 inch.
1-1/2 inch.
2-1/2 inch.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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Class B - Nonadjustable.

Size:

1 inch, with 1/4-inch tip 4-foot applicator.
1-1/2 inch, with 1/2-inch tip 10-foot applicator.
2-1/2 inch, with 1 inch tip 10-foot applicator.

Class C - Nonadjustable, high pressure, 1 inch.

Type III - Solid stream.

Class A - Plain.

Size:

1-1/2 inch, with 3/8-inch tip opening.
1-1/2 inch, with 1/2-inch tip opening.
1-1/2 inch, with 5/8-inch tip opening.
2-1/2 inch, with 1 inch tip opening.

Class B - Shut-off.

Size:

1 inch.
1-1/2 inch.
2-1/2 inch.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited).

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MILITARY

MIL-P-775	- Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-P-12164	- Playpipes: Nozzle, Fire Hose.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28/10	- Screw Thread Standards for Federal Services, Section 10, American National Hose Couplings and Fire-Hose Coupling Threads.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 47 - Malleable Iron Castings.

ASTM A 519 - Seamless Carbon and Alloy Steel Mechanical Tubing.

ASTM B 16 - Free-Cutting Brass Rod, Bar, and Shapes for Use in Screw Machines.

ASTM B 26 - Aluminum-Alloy Sand Castings.

ASTM B 108 - Aluminum-Alloy Permanent Mold Castings.

ASTM B 124 - Copper and Copper Alloy Forging Rod, Bar, and Shapes.

ASTM B 210 - Aluminum-Alloy Drawn Seamless Tubes.

ASTM B 584 - Copper Alloy Sand Castings for General Applications.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The nozzles shall provide either a continuous stream or a combination spray and stream under specified pressures and rate of flow. The nozzles shall be similar to the nozzles shown in figure 3 through 6 (see 6.5).

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The nozzle shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the nozzle may be exposed.

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3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Malleable cast iron. Malleable iron castings for nozzles shall conform to ASTM A 47, grade 32510.

3.3.3 Cast aluminum. Aluminum castings for nozzles shall conform to ASTM B 26 or ASTM B 108, Alloy UNS A03560, T6.

3.3.4 Brass.

3.3.4.1 Brass bars. Brass bars shall conform to ASTM B 16. Drop forged brass shall conform to ASTM B 124, Alloy UNS C37700.

3.3.4.2 Cast brass. Brass castings shall conform to ASTM B 584, Alloy UNS C83600 or UNS C84400.

3.3.5 Aluminum tubing. Aluminum tubing shall conform to ASTM B 210, Alloy 6061, temper to suit application.

3.3.6 Steel tubing. Steel tubing shall conform to ASTM A 519, type, grade and finish to suit application .

3.3.7 Gaskets. Gaskets shall be furnished on all nozzles equipped with internal threads.

3.4 Threads. All threads shall be American National Fire Hose Coupling Threads (NH) conforming to FED-STD-H28/10.

3.5 Design. The nozzles shall be aluminum, brass, malleable cast iron, or steel as specified (see 6.2). The nozzles shall have one end internally threaded for connecting to a fire hose or to a playpipe. Lugs, when provided, shall be the rocker-lug type.

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3.6 Environmental conditions.

3.6.1 Operating temperature. All moving parts of the nozzles shall be free moving and nonbinding under normal operating conditions in any ambient temperature from minus 25° F to plus 110° F.

3.6.2 Storage temperature. The nozzle assembly shall not be damaged by storage in any ambient temperature from minus 28° F to plus 145° F.

3.7 Hydrostatic pressure. The nozzles shall withstand an internal hydrostatic pressure of 500 pounds per square inch (psi), except that type II, class C nozzles shall withstand 1000 psi, without damage or physical distortion to the body of the nozzles which might impair its serviceability.

3.8 Finish. All ferrous surfaces, including threaded and sliding surfaces, shall be cadmium plated in accordance with QQ-P-416, class I, type II.

3.9 Identification marking. When specified (see 6.2), each nozzle shall be identified in accordance with MIL-STD-130. Marking shall be placed in a conspicuous location on the body of the nozzle.

3.10 Type I, revolving, cellar, 2-1/2 inch. The nozzle shall consist of a revolving barrel having not less than six machined orifices set at different angles, and not less than two water-stream orifices of different sizes so positioned as to cause the barrel to rotate when the nozzle is in operation. The barrel shall rotate on enclosed ball bearings which shall insure free-moving and nonbinding operation at a nozzle pressure of 100 psi. The revolving cellar nozzle shall be such shape and size as to pass through a 6-inch circular opening and shall be similar to figure 3 (see 6.5). The nozzle shall discharge not less than 280 gallons per minute (gpm) of water at a pressure of 100 psi and shall weigh not more than 9 pounds. The inlet connection shall have 2 1/2 - 7 1/2NH internal threads. The overall length of the nozzle shall be not less than 7 inches.

3.11 Type II, combination spray and stream.

3.11.1 Class A, adjustable. The adjustable nozzle shall have a control mechanism capable of producing a continuous stream varying from a solid stream to a conical fog pattern and an adjustable gallons flow mechanism to control water discharge. The nozzle tip shall be equipped with a natural rubber or synthetic rubber ring. The nozzle shall be of the weight and flow characteristics shown in table I.

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TABLE I. Weight and flow characteristics.

Internal thread size (inches)	Weight (brass) (pounds)		Flow characteristics (discharge pressure, 100 psi)	
			Straight stream (gpm)	Spray (fog) (gpm)
	Minimum	Maximum	Minimum	Minimum
1	2	4-3/4	12	12
1-1/2	2-1/2	6-3/4	95	95
2-1/2	4	11-1/2	120	120

3.11.1.1 Size 1-inch nozzle. The control mechanism shall operate smoothly at pressures up to 400 psi and shall provide a continuous change from straight stream to conical water fog pattern at any angle up to 140 degrees. The shutoff valve shall not leak more than 0.5 percent of specified minimum discharge at 100 psi. The butt end of the nozzle shall have 1 - 8NH internal threads.

3.11.1.2 Size 1-1/2 inch and 2-1/2 inch nozzle. The control mechanism shall operate smoothly at pressures up to 150 psi and shall provide a continuous change from straight stream to conical water fog pattern at any angle up to 140 degrees. The shutoff valve shall not leak more than 0.5 percent of specified minimum discharge at 100 psi. The butt end of the nozzles shall have 1 1/2 - 9NH and 2 1/2 - 7 1/2 NH internal threads, as applicable.

3.11.2 Class B, nonadjustable. The nonadjustable nozzle discharge end shall conform to figure 2 and as specified herein. The nozzle shall be similar to that shown in figure 4 (see 6.5). The nozzles shall be shutoff type and shall have two waterways between the shutoff valve and the tip. One waterway shall lead to a solid stream opening and the other to a removable spray-head opening. The design shall be such that there will be an enlarged water chamber in the spray-head-opening waterway which will lessen water turbulence and reduce friction loss. The valve mechanism shall be operated by means of a D-frame or horseshoe-type handle with ball-type stops at each of the following three positions: Beginning at the front, the positions shall be marked (1) "SHUT" (2) "FOG" or "SPRAY", (3) "OPEN" or "STREAM" in that order. In the "SHUT" position, there shall be no flow through the nozzle at the pressures specified; in the "FOG" or "SPRAY" position, water shall be delivered to the spray-head opening; in the "OPEN" or "STREAM" position, water shall be delivered to the solid-stream opening. The nozzle shall operate at pressures up to 500 psi under sea water conditions.

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3.11.2.1 Shutoff valve. The shutoff valve shall be full-flow, ball (single or double seat type), or cylindrical floating-gland type with ground joints. The floating gland shall operate by pressure through bypasses in the body of the ball or cylinder. The valve shall be leakproof under pressures up to 500 psi.

3.11.2.2 Tip. A tip of the size specified herein shall be incorporated in the solid-stream opening of the nozzle. The tip shall be smooth-bore type and shall be built into the nozzle to deliver a solid stream.

3.11.2.3 Attachments. Each nozzle shall be equipped with a removable, high-velocity spray head, attached to the main nozzle body by a metal chain, and a removable applicator with a low-velocity spray head. The applicator tube shall be aluminum tubing of the size and wall thickness shown in figure 2. The spray-head opening of the nozzle shall be constructed to receive either a high-velocity head or an applicator by means of a bayonet joint. The applicator shall have an additional seat behind the bayonet bushing seat in accordance with figure 2. Bends in the applicators shall have 6- to 8-inch radius and shall be downward in the vertical plane of the nozzle. The length of the applicator is the length of the tube before bending. The low-velocity spray head shall be fastened by means of external and internal threads to a bushing on the applicator. The low-velocity spray head shall be fitted with a strainer having an area 20 times the total effective free-water area through the orifices. The strainer openings shall measure not more than 3/64-inch. The strainer shall be designed to have a base that will push-fit in the head. The bushings for the bayonet joint and the head shall be pressure rolled or welded.

3.11.2.4 Sizes.

3.11.2.4.1 Size 1-inch nozzle. The tip for the 1-inch nozzle shall have a 1/4-inch opening. The high-velocity spray head shall deliver not less than 12 gpm of water fog in a consistent 90-degree conical pattern at a nozzle pressure of 100 psi, and deliver a straight stream of not less than 17 gpm at a nozzle pressure of 100 psi. The applicator shall be 4 feet long and shall be curved at an angle of 45-degrees at the spray-head end. An additional seat, 1-inch in diameter, shall be provided for that end of the applicator which fits into the nozzle body. If smaller than 1-inch outside diameter (OD) tubing is used for the applicator, the end that seats in the nozzle body shall be expanded or bushed so that it will fit properly. The applicator head shall be designed to produce an umbrella fog pattern and to discharge not less than 15 gpm of water at 100 psi. The butt end of the nozzle shall have 1 - 8NH internal threads.

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3.11.2.4.2 Size 1-1/2 inch nozzle. The tip for the 1-1/2 inch nozzle shall have a 1/2-inch opening. The straight bore and high-velocity spray head shall discharge not less than 65 gpm straight stream at 100 psi, and not less than 65 gpm of water fog in a consistent 70-degree conical pattern at a nozzle pressure of 100 psi. When ejected at this pressure, the spray shall project not less than 35 feet in a circular form with an additional drift of not less than 45 feet in still air. Breakup of the spray shall be uniform at pressures from 25 to 100 psi. The applicator shall be 10 feet long and shall be curved at an angle of 90-degrees at the spray-head end. If smaller than 1-1/4 inch OD tubing is used for the applicator, the end that seats in the nozzle body shall be expanded or bushed so that it will fit properly. The low-velocity head shall discharge not less than 54 gpm at 100 psi in an umbrella-fog-type pattern. The butt end of the nozzle shall have 1 1/2 - 9NH internal threads.

3.11.2.4.3 Size 2-1/2 inch nozzle. The tip for the 2-1/2 inch nozzle shall have a 1-inch opening. The straight bore and high-velocity spray head shall discharge not less than 132 gpm straight stream at 100 psi, and not less than 100 gpm nor more than 150 gpm of water fog in a consistent 70-degree conical pattern at a nozzle pressure of 100 psi. When ejected at 100 psi, the spray shall be projected not less than 35 feet in a circular form with an additional drift of not less than 45 feet in still air. Breakup of the spray shall be uniform at the above pressure. The applicator shall be 10 feet long and shall be curved at an angle of 90-degrees at the spray-head end. If smaller than 1-1/2 inch OD tubing is used for the applicator, the end that seats in the nozzle body shall be expanded or bushed so that it will fit properly. The low-velocity head shall discharge not less than 100 gpm at 100 psi in an umbrella-fog-type pattern. The butt end of the nozzle shall have 2 1/2 - 7 1/2 NH internal threads.

3.11.3 Class C, nonadjustable, high-pressure, 1-inch. The nozzle shall conform to the type II, class B, 1-inch nozzle, except that no applicator or low-velocity spray head shall be provided, and the nozzle shall discharge not less than 25 gpm at 800 psi for fog or straight stream. The fog pattern shall be a consistent cone with an included angle of not less than 90-degrees. The spray head shall be firmly attached to the main body by a metal chain. The valve shall be easily operated and shall leak not more than 10 drops per minute at 800 psi. The symbol "HP" shall be cast integrally on the main body of the nozzle to provide quick and easy identification for high-pressure use.

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3.12 Type III, solid stream.

3.12.1 Class A, plain. The class A nozzle shall be of one-piece construction as shown in figure 5 (see 6.5), and shall be not less than 10 inches long. The waterway bore shall be smooth and unobstructed and shall taper uniformly from the butt to the tip end. The discharge end of the bore shall be machined straight to proper size with smooth walls for a depth of not less than 1-inch, and shall be counterbored at the tip to a diameter 1/8-inch larger than the tip opening for a depth of 1/16-inch. The nozzles shall withstand hydrostatic pressures up to 500 psi. Tip openings, 3/8-inch, 1/2-inch, and 5/8-inch, shall be threaded 1 1/2 - 9 NH internal hose connection, and tip opening, 1-inch, shall be threaded 2 1/2 - 7 1/2NH internal hose connection.

3.12.2 Class B, shutoff. The class B nozzle shall have a built-in valve conforming to 3.11.2.1, except that the valve shall provide 1/4-turn shutoff and shall be fitted with a D-type or horseshoe handle. The nozzle body shall be a one-piece casting.

3.12.2.1 Size 1-inch nozzle. The nozzle shall be as shown in figure 6 (see 6.5). The diameter of the waterway shall be not less than 1/2-inch. The inlet connection for the hose shall have 1 - 8NH internal threads, and the outlet connection for the tip shall have 3/4 - 8NH external threads.

3.12.2.2 Size 1-1/2 inch nozzle. The nozzle shall be as shown in figure 6 (see 6.5). The diameter of the waterway shall be not less than 3/4-inch. The inlet connection for the hose shall have 1 1/2 - 9NH internal threads, and the outlet connection for the tip shall have 1 - 8NH external threads. When specified (see 6.2), the tip end of the nozzle shall have 1 1/2 - 9NH external threads.

3.12.2.3 Size 2-1/2 inch nozzle. The nozzle shall be as shown in figure 6 (see 6.5). The diameter of the waterway shall be not less than 1-1/4 inches. The butt end of the nozzle shall have a 1 1/2 - 9NH internal thread connection. The tip end of the nozzle shall have a 1 1/2 - 9NH external thread. When specified (see 6.2), the nozzle shall be furnished with a playpipe conforming to MIL-P-12164, Type II.

3.13 Workmanship. All parts, components, and assemblies of the nozzles which include castings and forgings shall be clean from harmful extraneous material such as sand, dirt, sprues, and scale. Surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges of corners are required or where they are not detrimental to safety or function of the nozzles.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

(c) Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article nozzle shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article nozzle shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.1.2 Tests. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined for the defects specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1.2 shall be tested as specified in 4.5.2.3. AQL shall be 2.5 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. The nozzle shall be examined as specified herein for the following major defects:

101. Material not as specified.
102. Materials are not corrosion and deterioration resistant or treated to be made corrosion and deterioration resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Rocker lugs not as specified.
106. Threads not as specified.
107. Components missing or not as specified.
108. Finish not as specified.
109. Identification marking missing, incomplete, or illegible.
110. Symbol "HP" missing from type II, class C nozzle.
111. Weight and length not as specified for type I nozzle.
112. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Environmental.

4.5.2.1.1 High temperature. The nozzles shall be placed in a test chamber and the temperature of the chamber raised. The maximum ambient storage temperature shall 145° F, and the maximum ambient operating temperature shall be 110° F with a relative humidity of not more than 15-percent. The temperature shall be maintained for not less than 4 hours. While at 110° F the moving parts of the nozzles shall be operated. Sticking, binding, or damage shall constitute failure of this test.

4.5.2.1.2 Low temperature. The nozzles shall be placed in a test chamber and the temperature of the chamber lowered. The minimum ambient storage temperature shall be minus 28° F and the minimum operating temperature shall be minus 25° F. The temperature shall be maintained for not less than 4 hours. While at minus 25° F the moving parts of the nozzles shall be operated. Sticking, binding or damage shall constitute failure of this test.

4.5.2.2 Hydrostatic pressure. The nozzles shall be subjected to an internal hydrostatic pressure of 500 psi except that type II, class C nozzles shall be subjected to 1000 psi. Apply pressure with the valves open and the tips plugged on adjustable and shutoff-type nozzles, so that the entire internal area of the nozzles is subjected to the pressure specified. Nonconformance to 3.7 shall constitute failure of this test.

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4.5.2.3 Discharge capacity. The discharge capacity of the nozzle is determined by either of the following two methods: Method 1 is by attaching a calibrated pressure gage (psi) at the effluent end of the nozzle and a calibrated flowmeter (gpm) between the influent end of the nozzle and the water feedline. Method 2 is by attaching a claibrated pressure gage at the effluent end of the nozzle and by collecting the total discharge in a calibrated tank in a unit of time. Nonconformance to 3.10, 3.11.1, 3.11.2.4, or 3.11.3 shall constitute failure of this test.

4.5.2.4 Fog pattern. Conduct the fog pattern test on the type II nozzles only.

4.5.2.4.1 Procedure. Determine the fog pattern angle and effective discharge range by placing the nozzle not less than 6 feet in front of a backboard grid as shown in figure 1 (see 6.5). Nozzle pressure shall be determined by a pressure gage placed in the water feedline immediately behind the nozzle. The pattern against the backboard shall be photographed, and the angle and range of the pattern shown on the photograph shall be determined. Failure of the fog pattern angle and range to comply with 3.11.1, 3.11.2.4.1, 3.11.2.4.2, 3.11.2.4.3, and 3.11.3 shall constitute failure of this test.

4.5.2.5 Bayonet joint. To insure a positive fit between the nozzle and the applicator, insert the applicator into each nozzle and subject the nozzle and the applicator to a hydrostatic water pressure of 100 psi. Any leakage in excess of 10 cubic centimeters per hour shall constitute failure of this test.

4.5.2.6 Ball valve leakage. Subject the valves to a hydrostatic water pressure of 500 psi in the shut position. Any leakage in excess of 10 cubic centimeters per hour shall constitute failure of this test.

4.5.2.6.1 High-pressure-ball-valve leakage. Subject the high-pressure-nozzle shutoff valve to a hydrostatic water pressure of 800 psi. Any leakage in excess of 20 cubic centimeters per hour shall constitute failure of this test.

4.6 Inspection of packaging. Preservation, packing and marking for level A or B shall be inspected in accordance with the Quality Assurance Provisions of MIL-P-775. Commercial preservation, packing, and marking shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

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5.1.1 Level A. The nozzles shall be preserved in accordance with the level A preservation and packaging requirements of MIL-P-775.

5.1.2 Commercial. The nozzles shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. The nozzles, preserved as specified in 5.1, shall be packed in accordance with the level A packing requirements of MIL-P-775.

5.2.2 Level B. The nozzles, preserved as specified in 5.1, shall be packed in accordance with the level B packing requirements of MIL-P-775.

5.2.3 Commercial. The nozzles, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The nozzles are intended to be attached to the end of a fire hose to direct a stream of water used in combating fires.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class and size required (see 1.2).

c. When a first article is required for inspection and approval, and number of nozzles required (see 3.2).

d. Material required (see 3.5).

e. When marking shall be in accordance with MIL-STD-130 (see 3.9).

f. When 1 1/2 - 9NH external threads are required (see 3.12.2.2).

g. When playpipe is required (see 3.12.2.3).

h. Degree of preservation and degree of packing required (see 5.1 and 5.2).

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first produced nozzle. The first article should consist of one or more nozzles. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, tests, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Information figures. Figure 1, and figure 3 through 6 show a type of nozzle which has been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of another nozzle which conforms to this specification.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review activity:
Army - CE
DLA - CS
Air Force - 84

User activities:
Navy - MC

Preparing activity:
Army - ME

Project 4210-0353

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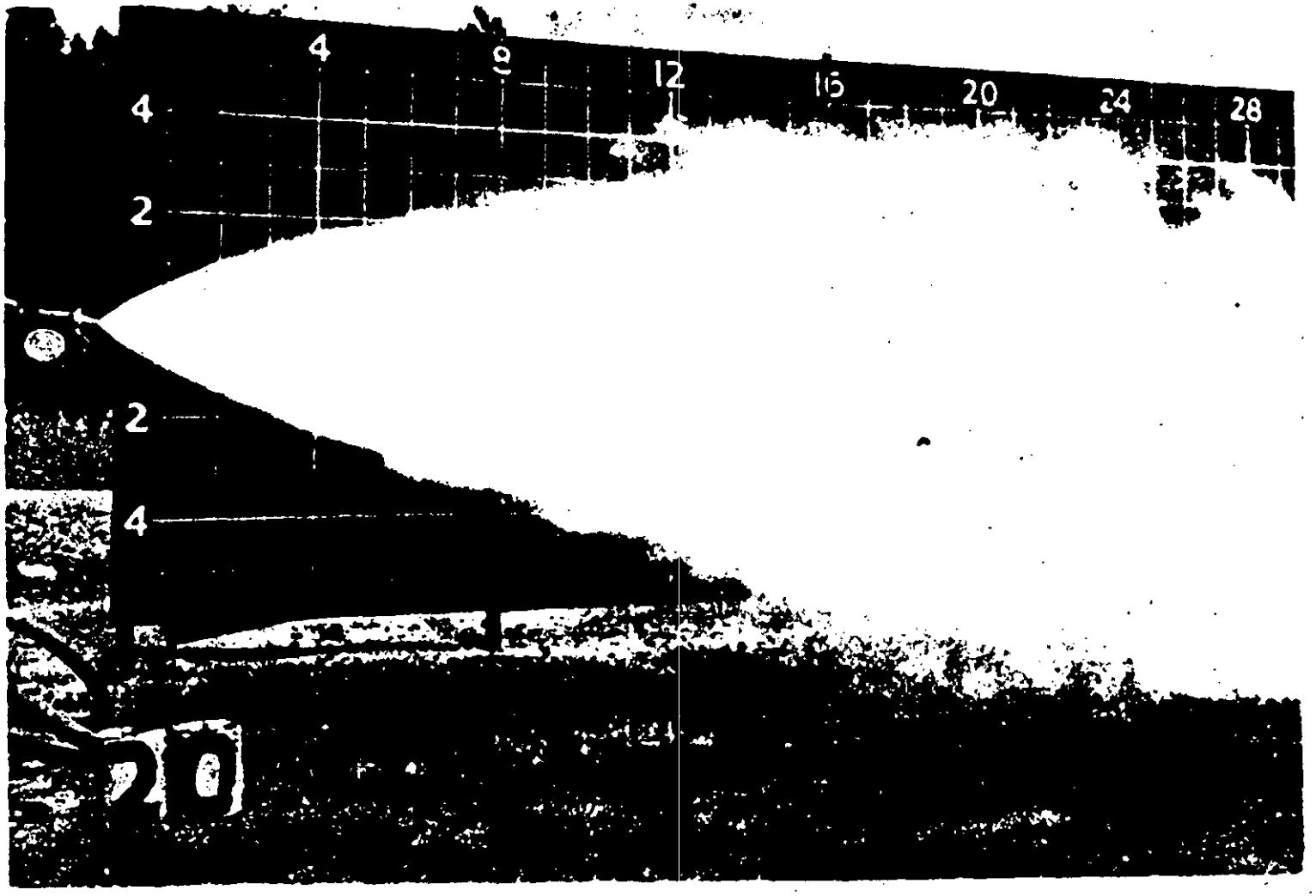
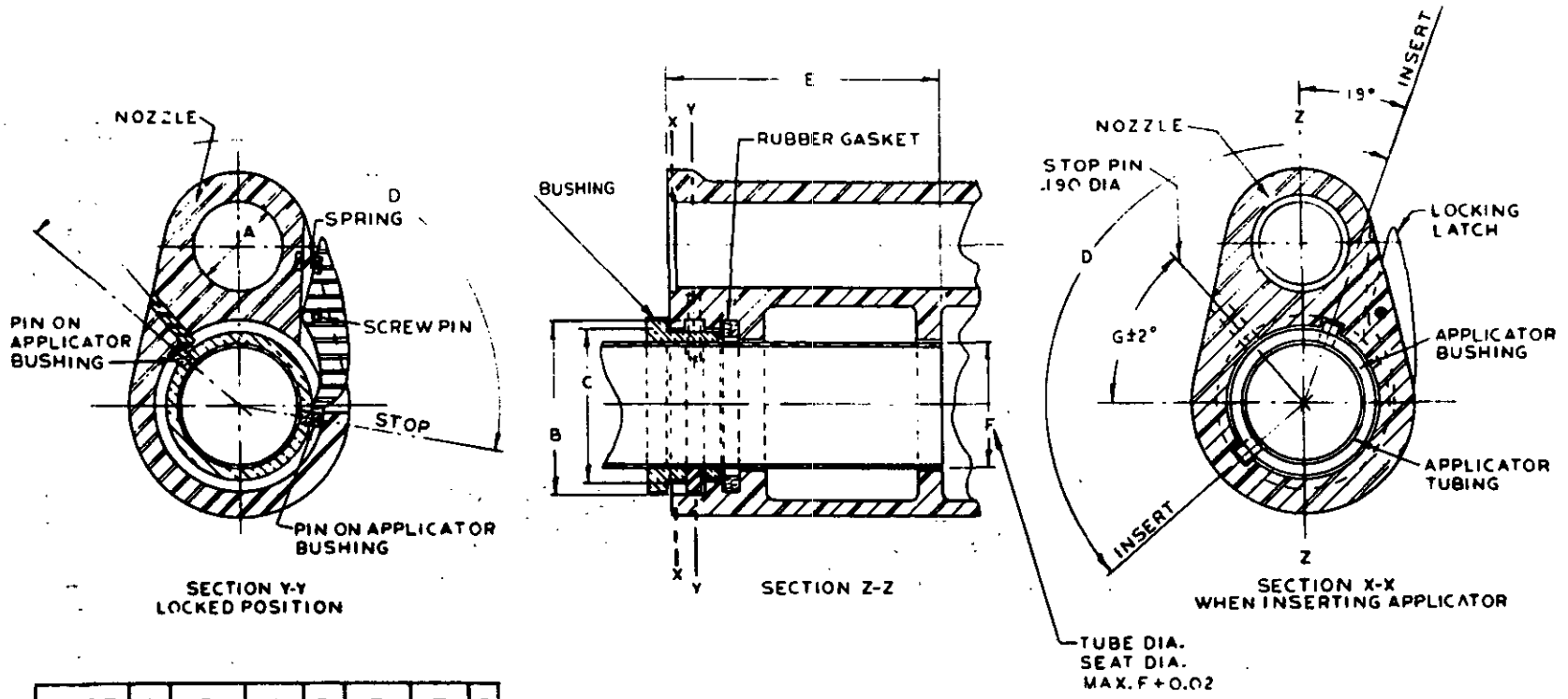


FIGURE 1. Water fog nozzle test using backboard grid.

X-1626 A

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SIZE	A	B	C	D	E	F	G
1	.25	1.38	1.19	150°	2.50	1.00	54°
1 1/2	.50	1.91	1.75	150°	3.28	1.25	54°
2 1/2	1.00	1.91	1.75	165°	3.50	1.50	38°

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FEATURES NOT DIMENSIONED ARE OPTIONAL.
3. APPLICATOR TUBING WITH 1-INCH OR LESS OUTSIDE DIAMETER SHALL HAVE WALL THICKNESS OF .049-INCHES. APPLICATOR TUBING WITH OUTSIDE DIAMETER GREATER THAN 1-INCH SHALL HAVE A WALL THICKNESS OF .065-INCHES. OUTSIDE DIAMETER OF APPLICATOR TUBING MAY BE REDUCED AS MUCH AS .25-INCH BELOW THAT SPECIFIED.

FIGURE 2. Nozzle, type II, class B, 1-inch, 1 1/2-inch, 2 1/2-inch sizes.

X-1359B

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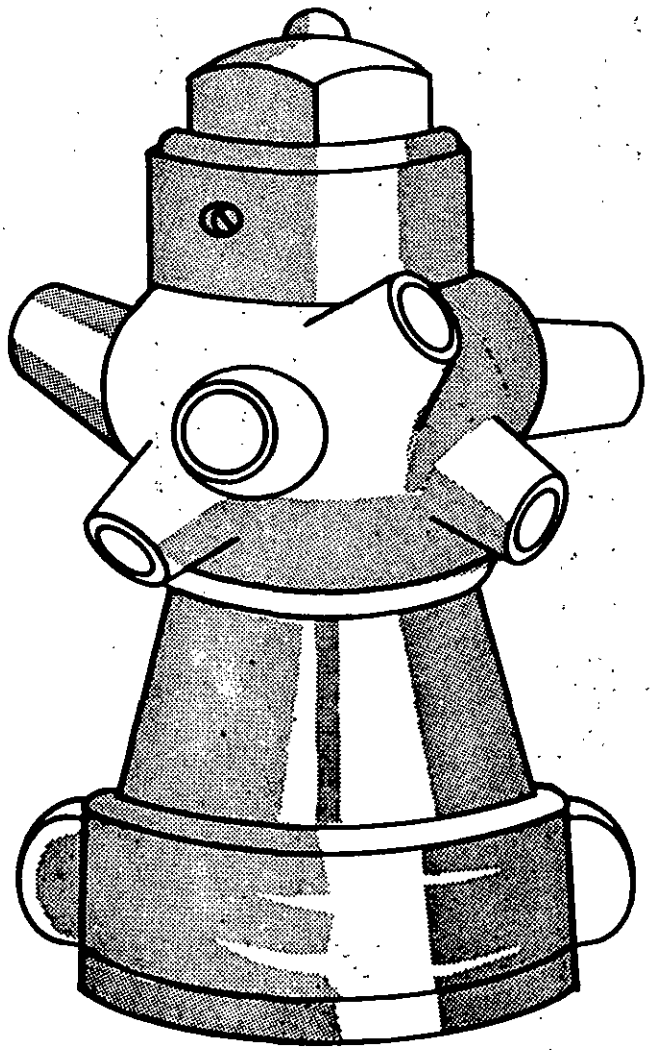


FIGURE 3. Type I.

X-1360A

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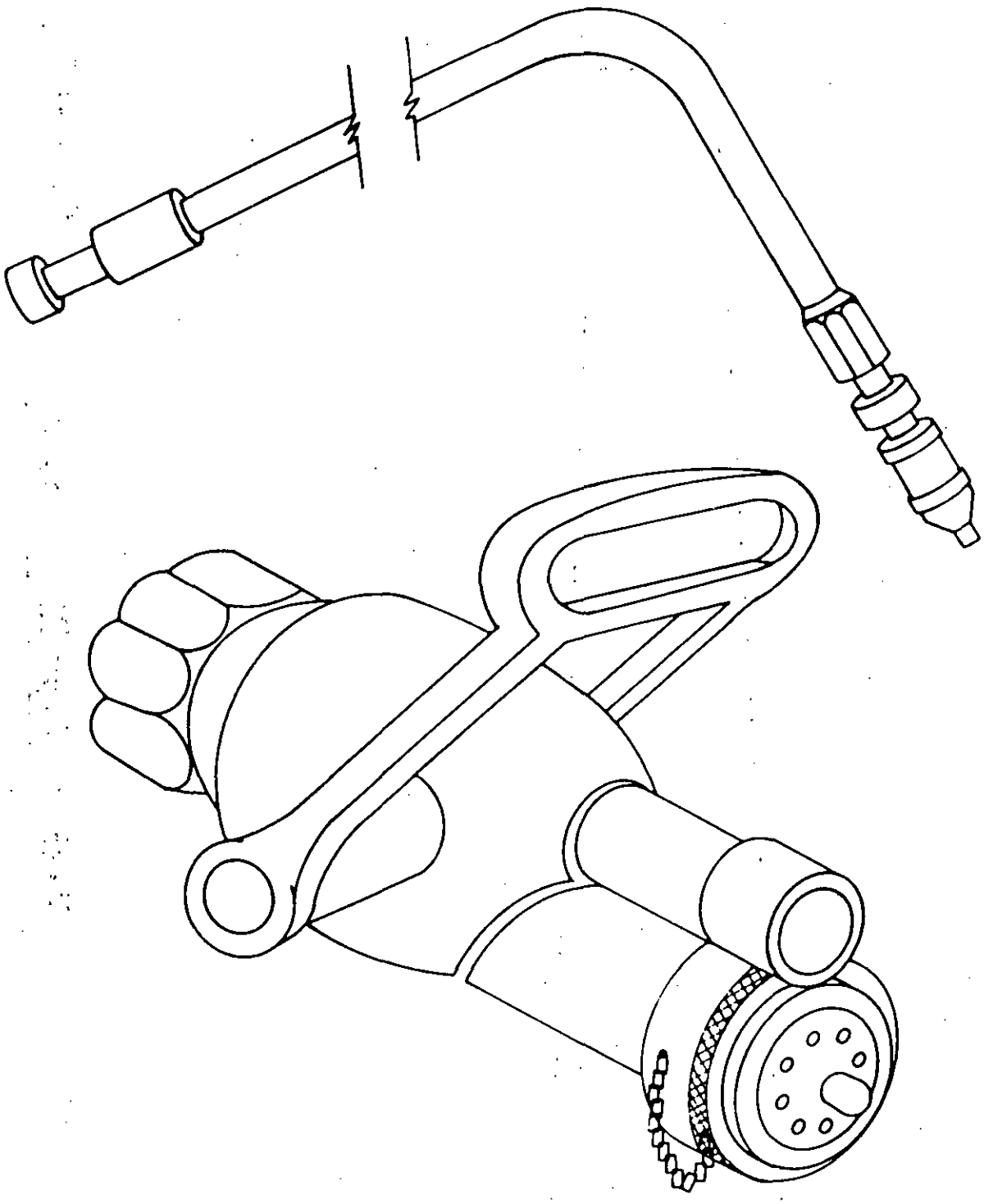


FIGURE 4. Type II, class B. X-3889A

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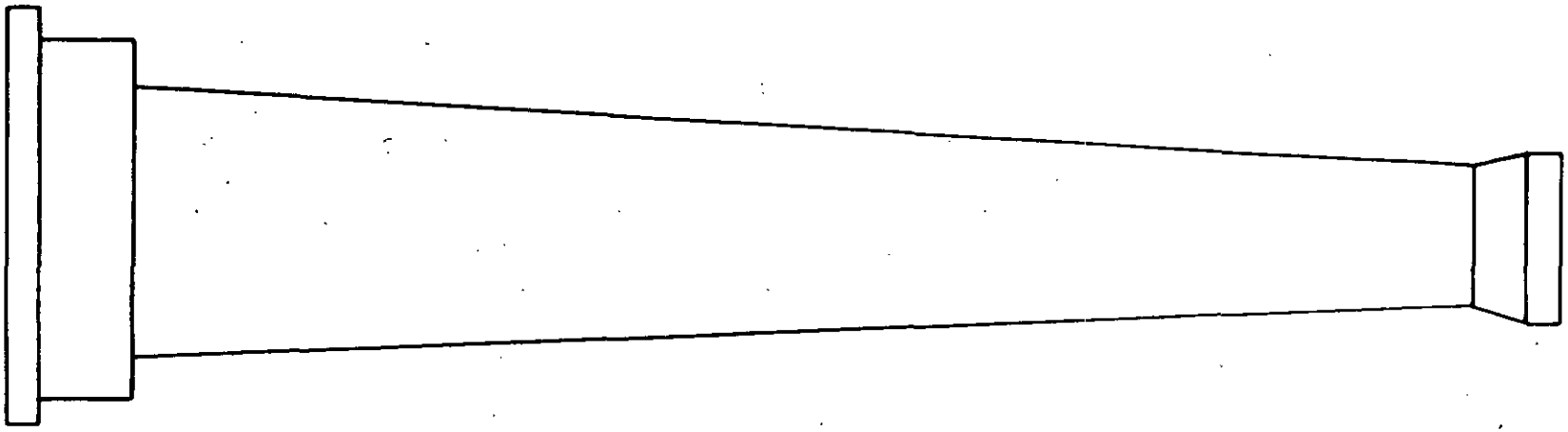


FIGURE 5. Type III, class A.

X-3890A

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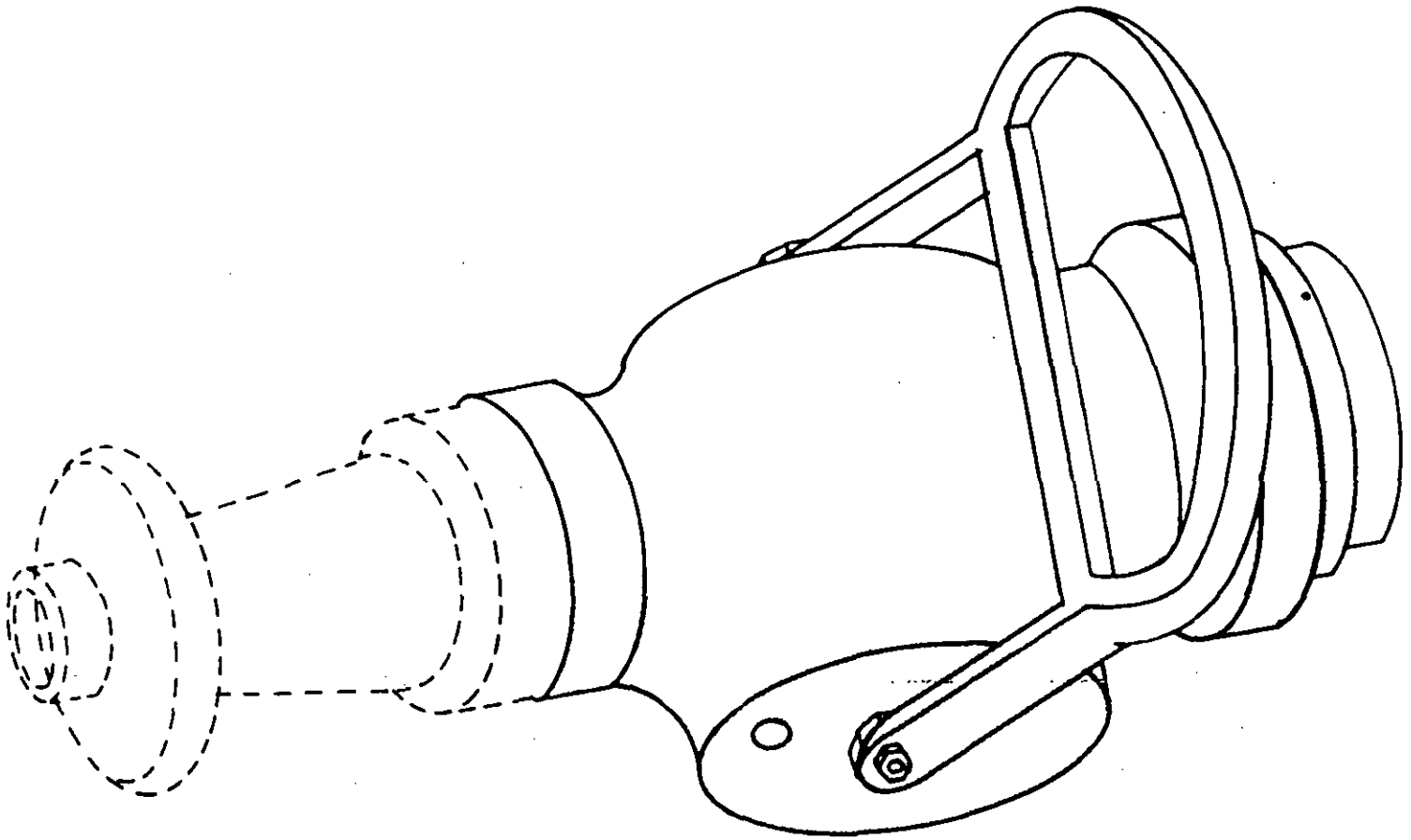


FIGURE 6. Type III, class B.

X-3891A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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